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Palaeomagnetism of the Serra Geral Formation

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Summary

Measurements are described of the directions of magnetization of samples of basaltic flows and diabase dikes and sills from the Serra Geral Formation at 33 sites in Brazil and Uruguay. The precision of the data is improved after treatment in alternating magnetic fields. Both normal and reversed magnetization were found in the basaltic lava flows and in the diabase dikes and sills. The angle between average normal and reversed directions is not significantly different from 180° and the mean direction of magnetization of the lava flows is not significantly different from that of the dikes and sills. Thermo-magnetic analysis shows the magnetic constituent to be magnetite. The susceptibility and intensity of magnetization are also discussed. The scatter of the sample-mean and site-mean directions of magnetization is analysed and the latter attributed principally to secular variation of the palaeogeomagnetic field.

1. Introduction

Results of a preliminary palaeomagnetic survey of the basalts and diabases of the Serra Geral formation were described by Creer (1958), who presented measurements from sites in Uruguay at the southern extremity of the region where this formation is exposed.

In this paper are presented data from 80 samples of rock from 33 sites in Southern Brazil and Uruguay. These 80 samples were cut into about 250 cylindrical specimens which were measured in the usual way under an astatic magnetometer.

2. Geology of the Serra Geral Formation

A brief account of the Serra Geral Formation is given by de Oliveira (p. 44 of the Handbook of South American Geology, Geological Society of America Memoir 65, 1956) and a more complete account by Leinz (1949).

The Serra Geral Formation constitutes the major known mass of volcanic rocks in the world, covering a surface area of about $10^6 km^2$ in the states of São Paulo, Paraná, Santa Catarina and Rio Grande do Sul in Southern Brazil, and also in Uruguay, (see Figure 1). About half is overlain by younger sediments.

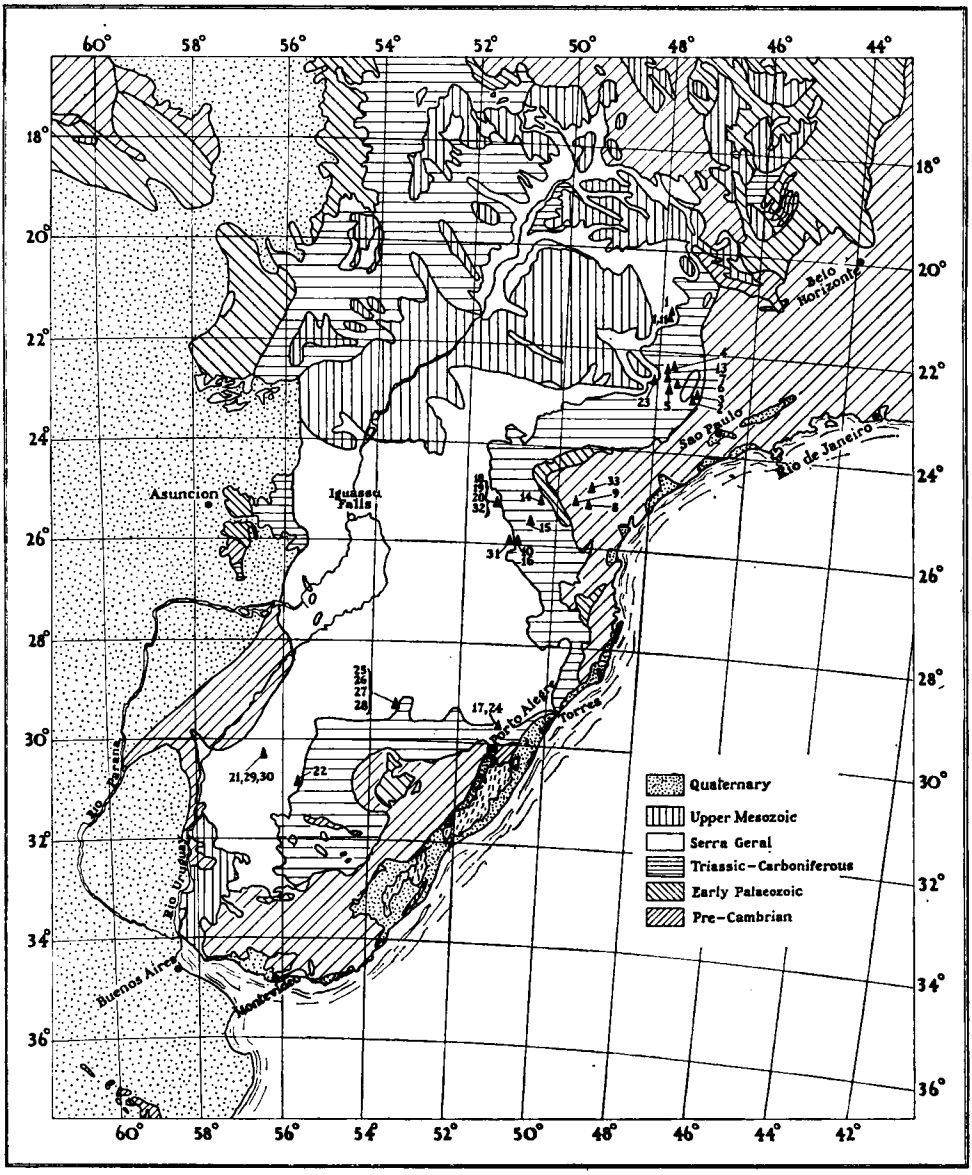


FIG. 1.—Geological environment of Serra Geral Formation.

The flows are about 400 m thick over a large part of this area, though local thicknesses of about 800 m are known, and a maximum of 1 000 m has been measured near Torres, Rio Grande do Sul. The total volume of lava flows is estimated at about $3{\cdot}5 \times 10^5 km^3$.

Up to 13 flows have been recognized in profile and they are locally separated by lenticular layers of cross-bedded aolean sandstone, of thickness up to 40 m,

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particularly in the north of the area. In the south, as intercalated sandstones are not always present, the existence of separate flows is often recognized by vesicle and vitreous textures together with horizontal diaclases on the top and bottom of each flow. Individual flows vary in thickness up to 80 m.

Associated with the lava flows, particularly in the region surrounding them are a number of steeply dipping fault planes, some of which are filled by diabase dikes which vary in thickness from a few metres to tens of metres. These cut the underlying formations and have also been observed intruding into the eruptives so that a later age has been assigned to the dikes than to the rocks of the basal flow. Diabase sills also occur, and some of these are about 200 m thick and cover areas of hundreds of km^2.

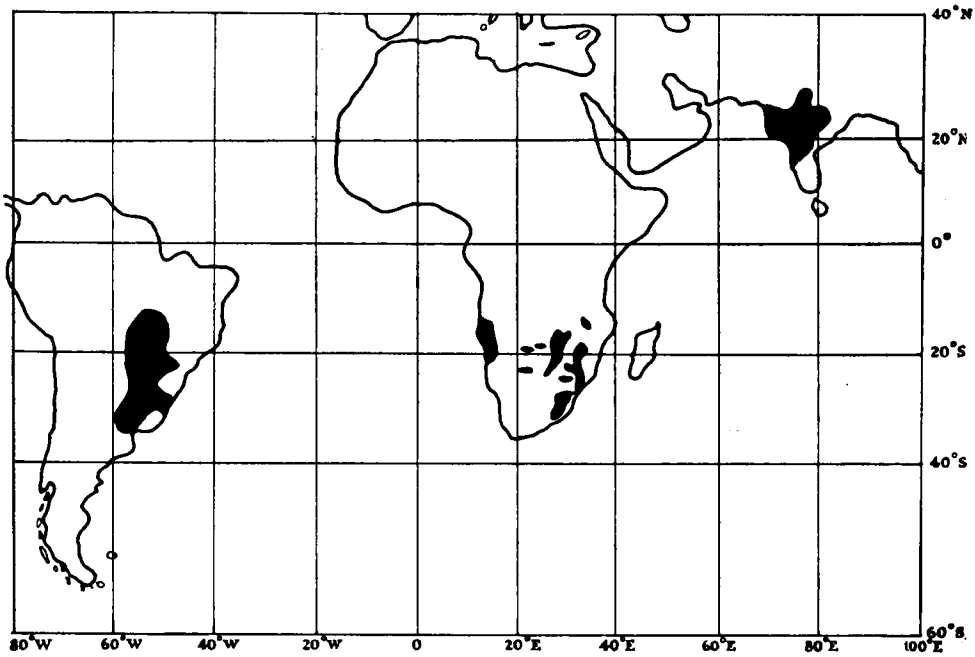


FIG. 2.—Geographical extent of the Mesozoic basalts and diabases of the southern hemisphere.

The chemical composition of the basaltic magma is said to be similar to that of the magmas of other Gondwanic flows. The Serra Geral formation rests unconformably on the Botucatú Sandstone which is unfossiliferous as a rule. Near Santa Maria in Rio Grande do Sul the basal flow occupies a valley cut down to within 19 m of a bed of fossil bones in the Santa Maria formation which is dated from the base of the Upper Triassic. However the thick Botucatú Sandstone and two unconformities separate the Serra Geral Formation from the dated Santa Maria Formation so the age of the former can only be placed between the Upper Triassic and Upper Cretaceous, possibly between the Rhaetic and Jurassic.

The geographical extent of the Serra Geral Formation is shown in Figure 2 together with that of the Karroo Formation of South Africa and the Deccan Traps of India.

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3. The measurements

The directions of magnetization of cylindrical specimens ($\frac{3}{4}$ in. in diameter and $\frac{3}{4}$ in. long) were measured and those from a given sample of rock (usually 3 cylinders per sample) were averaged and the directions of magnetization of each of the 80 samples are shown in a stereographic projection in Figure 3a.

It has previously been shown (Creer (1958)) that some rocks from the Serra Geral Formation have acquired a secondary magnetization which masks the directions of the fossil thermoremanence and that such a secondary magnetization can be destroyed by treatment in alternating magnetic fields. The cylindrical specimens were therefore so treated (peak field value 250 Oe), and after this the directions of magnetization from rocks from 30 of the 33 sites could be classified into two well-defined groups roughly 180° apart. Following usual practice, the group of directions close to the present field is called "normal" and the other "reversed". In Figure 3b the directions of magnetization after "cleaning" are shown for the 74 samples which could be so classified. The "cleaned" directions of magnetization of samples from each site were then averaged and the 30 site directions of magnetization are shown in Figure 3c. In both Figures 3b and 3c directions from basaltic flows are distinguished from those from diabase dikes and sills.

The geographical positions of the sites are shown in Figure 1 and also in Table 1 (column I). The mean directions of magnetization of the rock samples before and after cleaning are given in column II and the average directions of magnetization for sites after cleaning are given in column III. In columns IV and V are given the positions of the north poles corresponding to the listed directions of magnetization assuming a dipole field, and calculated as described in Creer, Irving & Runcorn (1957).

4. Classification and analysis of results

The samples and sites have been classified into four groups, *viz.* normal and reversed basaltic flows and normal and reversed diabase dikes or sills. Average directions of magnetization (D and I) for each of these groups are given in Table 2, where sample directions have been assigned unit weight and in Table 3 where site directions have been assigned unit weight. The number of unit vectors (N) is given in these tables and the length of the resultant vector (R) is listed so that any statistical quantity not given here may be calculated. The statistics K, α, θ_0 and θ_e are also given. K defines the precision of the data and has been widely used in palaeomagnetism, being a parameter defining the width of a Fisherian distribution in which the density of observed directions making an angle θ with the mean direction varies like $\exp(K\cos\theta)$. Fisher (1953) has shown that the maximum likelihood estimator of K is given by $(N-1)/(N-R)$, K being large when the data are tightly grouped about the mean and equal to 1 for a random distribution. θ_0 is the angle in degrees equivalent to the standard deviation σ in a Gaussian distribution and is given by $\theta_0 = (2/K)^{\frac{1}{2}}$ radians $= (81/K)^{\frac{1}{2}}$ degrees. It is very nearly the same as the analogue of standard deviation $\delta = \cos^{-1}R/N$ used by Wilson (1959) except for small values of N. θ_e is the angle in degrees equivalent to the standard error of a Gaussian distribution and is obtained from $\theta_e = \theta_0/N^{\frac{1}{2}}$. The angle α is the semi-angle of the cone of confidence drawn at the 95 per cent probability level. It varies like θ_e but is always smaller because

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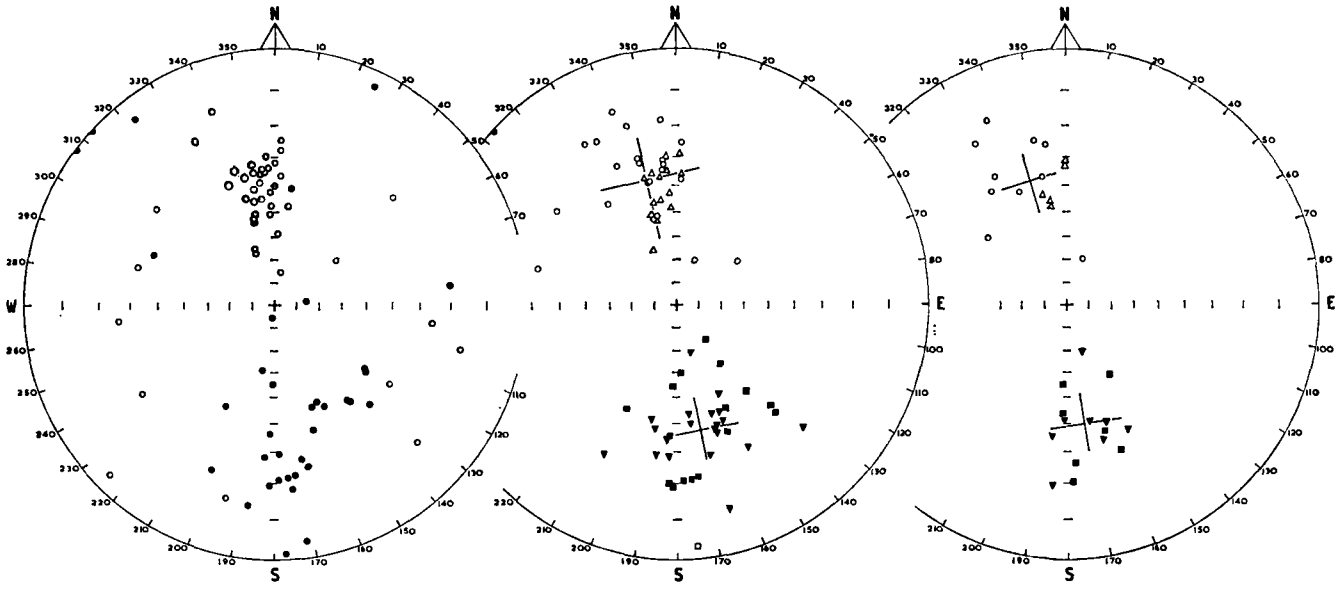


FIG. 3.—Directions of magnetization:

(a) Mean directions of magnetization of 80 hand specimens before cleaning. Each direction is the mean of about three small cylindrical specimens cut from each hand specimen. (b) Mean directions of 74 hand specimens which could be classified as "normal" or "reversed" after "cleaning". (c) Mean directions of magnetization of 30 sites.

Key to figures 3b and 3c:

○ normally magnetized diabase; △ normally magnetized basalt;

■ reversely magnetized diabase; ▼ reversely magnetized basalt.

Hollow symbols represent directions in upper hemisphere (up); solid black symbols represent directions in lower hemisphere (down).

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Table 1

No.	Name	I PARTICULARS OF SITE Longitude	Latitude	II DIRECTIONS OF MAGNETIZATION SAMPLES (Original)	SAMPLES (Cleaned)	III SITES (Cleaned)	IV NORTH POLE POSITIONS SAMPLES Longitude	Latitude	V SITES Longitude	Latitude
				I NORMALLY MAGNETIZED DIKES OR SILLS						
1	Riberão Preto	47°·7 W	21°·5 S	345° −52°	345° −52°	309°·6 −45°·5	179° E	73° N	155°·0 W	44°·0 N
				285° −31°	285° −31°		149° W	20° N		
2	Chapadas Quarry (near Campinas S.P.)	47°·2 W	22°·5 S	0° −32°	354° −34°	349°·0 −36°·5	104° W	83° N	127°·1 W	79°·5 N
				350° −32°	345° −30°		115° W	74° N		
				342° −33°	336° −29°		124° W	66° N		
				356° −51°	348° −51°		179° W	76° N		
				0° −40°	3° −37°		10° E	87° N		
3	Antonio Gabinato Quarry (near Limeira)	47°·5 W	22°·5 S	2° −37°	355° −18°	351°·7 −26°·4	68° W	76° N	91°·5 W	78°·4 N
				354° −34°	355° −32°		91° W	83° N		
				352° −39°	345° −29°		114° W	74° N		
4	Railway cutting between Batovi and Camacuá (Municipio Rio Claro—km 146·2 from Jundiai)	47°·5 W	22°·3 S	355° −35°	348° −37°	336°·7 −38°·6	131° W	79° N	140°·5 W	68°·4 N
				338° −37°	325° −39°		144° W	58° N		
5	Piracicaba (waterfall in river)	47°·7 W	22°·7 S	177° + 1°	346° −18°	349°·2 −24°·1	95° W	71° N	95°·0 W	75°·6 N
				291° −35°	355° −30°		84° W	82° N		
6	Paúlista Railway Co. Quarry, Tatú	47°·3 W	22°·6 S	349° −41°	347° −37°	326°·5 −34°·0	131° W	78° N	136°·4 W	58°·5 N
				308° −28°	308° −28°		138° W	40° N		
7	Sill 16 km from Rio Claro on road to Piracicaba	47°·5 W	22°·5 S	344° −44°	330° −18°	330° −18°	118° W	58° N	117°·6 W	58°·3 N
8	Dike near Bocaiura do Sul (Paraná)	49°·5 W	25° S	338° −66°	24° −69°	24° −69°	103° E	57° N	103°·4 E	57°·1 N
9	Gava Quarry, São Lourenço, (Paraná)	49°·5 W	25°·5 S	333° −19°	333° −19°	330°·1 −30°·1	113° W	60° N	126°·9 W	60°·6 N
				305° −57°	305° −57°		69° W	42° N		
				341° −13°	341° −13°		97° W	64° N		
10	Serra do Tigre, Mallet, Paraná	51° W	26° S	313° 0°	313° 0°	336°·2 −13°·6	119° W	37° N	105°·1 W	60°·3 N
				2° −25°	2° −25°		42° W	77° N		

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II REVERSELY MAGNETIZED DIKES OR SILLS

11	Quarry at Riberão Preto, S.P.	47°·7 W	21°·5 S	174° +18° 141° +38°	181° +20° 137° +30°	159°·9	+26°·7	43° E 138° W	79° N 49° N	119°·9 W	69°·5 N
12	Quarry at Riberão Preto, S.P.	47°·7 W	21°·5 S	2° +36° 172° +22° 172° +21°	182° +36° 172° +22° 172° +21°	177°·1	+26°·4	3° E 86° W 55° W	88° N 77° N 80° N	68°·2 W	82°·0 N
13	Quarry on Fazenda Santana da Cima, 27 km from Rio Claro, S.P.	47°·5 W	22°·5 S	322° + 5° 307° 0° 124° −32°	137° +70° 158° +34° 143° +58°	149°·1	+54°·4	167° E 131° W 173° W	45° N 69° N 55° N	169°·2 W	60°·3 N
14	Near Ponta Grossa, Paraná	50°·5 W	25° S	136° +33° 134° +42° 206° +42°	136° +33° 134° +42° 206° +42°	182°·0	+44°·1	140° W 149° W 44° E	49° N 48° N 66° N	65°·2 E	88°·0 N
15	7 km from Irati on road to Prudentopolis, Paraná	50°·7 W	25°·5 S	182° +55°	182° +55°	182°	+55°	120° E	80° N	120°·0 E	79°·8 N
16	Serra Rio Claro, Mallet Paraná	51°·0 W	26°·0 S	172° + 4° 125° +43° 191° +60°	175° − 3° 140° +43° 176° +60°	163°·9	+35°·0	62° W 148° W 141° E	62° N 54° N 75° N	120°·0 W	73°·7 N
17	Km 20, Federal Road to Rio de Janeiro, Porto Alegre, R.G. do S.	51° W	29°·6 S	175° +21° 181° +19°	175° +21° 181° +19°	178°·0	+20°·0	66° W 48° W	71° N 70° N	56°·9 W	70°·6 N

III NORMALLY MAGNETIZED FLOWS

18	Escarpment on road to Guarapava, near Prundentopolis, Paraná	51°·2 W	25° S	353° −36° 346° −36° 338° −64° 350° −35°	355° −36° 346° −36° 338° −64° 350° −35°	347°·9	−42°·7	95° W 123° W 164° W 111° W	83° N 76° N 63° N 79° N	142°·6 W	79°·0 N
19	As site 18, but lower stratigraphically	51°·2 W	25° S	2° −28° 357° −34°	2° −28° 357° −34°	359°·6	−31°·0	40° W 75° W	80° N 83° N	53°·9 W	81°·7 N
20	As site 19, but lower stratigraphically	51°·2 W	25° S	356° −30° 9° −76°	356° −30° 3° −36°	359°·4	−33°·0	75° W 22° W	80° N 84° N	55°·9 W	83°·0 N
21	Bank of R. Cuareim near Artigas, Uruguay	56°·5 W	30°·5 S	357° −43° 354° −52°	357° −43° 345° −52°	351°·5	−47°·7	83° W 160° W	84° N 77° N	135°·6 N	82°·4 N
22	Quarry of the Cerro del Marco, near Livramento, R.G. do S.	55°·5 W	31° S	358° −48° 352° −45° 346° −51° 348° −45°	358° −48° 352° −45° 346° −51° 348° −45°	351°·0	−47°·3	97° W 115° W 152° W 126° W	87° N 82° N 78° N 79° N	129°·7 W	81°·8 N

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Table 1—*continued*

No.	Name	I PARTICULARS OF SITE Longitude	Latitude	II DIRECTIONS OF MAGNETIZATION SAMPLES (Original)	SAMPLES (Cleaned)	III SITES (Cleaned)	IV NORTH POLE POSITIONS SAMPLES Longitude	Latitude	V SITES Longitude	Latitude
				IV REVERSELY MAGNETIZED FLOWS						
23	Serra Soã Pedro, Fazenda São João, 11 km from São Pedro, S.P.	48° W	22°·5 S	201° +20° 160° +44°	190° +38° 173° +44°	181°·9 +41°·3	37° E 166° W	81° N 83° N	77°·2 E	87°·9 N
24	Km 20 from Porto Alegre, R.G. do S. on Federal Road	51° W	29°·6 S	178° +29° 184° +28° 163° +36°	154° +48° 183° +35° 163° +36°	168°·9 +40°·3	146° W 35° W 113° W	67° N 79° N 72° N	109°·9 W	78°·1 N
25	Escarpment on road to Julho de Castilhos, near Santa Maria, R.G. do Sul	53°·5 W	29°·5 S	142° +39° 157° +45°	153° +35° 152° +42°	162°·5 +38°·5	127° W 138° W	64° N 64° N	121°·2 W	72°·4 N
26	As 25, but lower in succession	53°·5 W	29°·5 S	188° +13° 192° +83°	183° +29° 192° +41°	187°·2 +35°·1	41° W 10° E	76° N 78° N	19°·0 W	77°·9 N
27	As 26, but lower in succession	53°·5 W	29°·5 S	194° −14° 126° +42°	165° +11° 206° +24°	184°·7 +18°·6	87° W 7° E	62° N 60° N	40°·1 W	69°·6 N
28	As 27, but lower in succession	53°·5 W	29°·5 S	168° +24° 170° +27° 48° −26°	161° +36° 173° +40° 133° +20°	154°·3 +33°·2	119° W 98° W 130° W	70° N 81° N 42° N	123°·4 W	64°·0 N
29	4 km from Artigas, Uruguay on road to Rivera	56°·5 W	30°·5 S	8° −48° 85° +30° 84° +76° 236° −25° 25° + 4°	167° +28° 158° +38° 158° +41° 188° +29° 153° +26°	165°·0 +33°·0	97° W 127° W 132° W 28° W 118° W	70° N 68° N 69° N 73° N 60° N	107°·4 W	71°·5 N
30	Quarry near Artigas	56°·5 W	30°·5 S	8° +41°	163° +68°	163° +68°	150° E	66° N	150°·2 E	65°·9 N

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V OBLIQUELY MAGNETIZED SITES

31	Serra Rio Claro, Mallet Paraná	51° W	26° S	104° −16° 97° −26°	104° −16° 97° −26°	100°·6 −21°·0	53° E 54° E	9° S 0° N	53°·4 E	4°·6 S
32	Escarpment on road to Guarapava, near Prudentopolis	51°·2 W	25° S	224° − 4° 264° −26°	224° − 4° 264° −26°	242°·9 −15°·9	167° W 156° W	39° S 1° N	161°·1 W	20°·4 S
33	Ponte do Bariqui near Rio Branco do Sul, Paraná	49° W	25° S	21° −59° 133° −14°	107° −42° 133° −42°	121° −43°	70° E 84° E	4° S 23° S	77°·8 E	14°·2 S

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θ_e is the semi-angle of a similar cone within which, for a Fisherian distribution, the probability that the true mean lies is 63 per cent (see Creer, Irving & Nairn (1959)).

Table 2

Average Directions of Magnetization

Sample Directions Assigned Unit Weight

D	I	N	R	K	α	θ_0	θ_e
			All Original Data				
+354·1	−59.3	+80·0	+11·0009	+1·1	+40·7	+75·7	+8·5
			Original Data—Apparently Normal Samples				
+347·4	−42·4	+46·0	+36.9651	+5.0	+10·5	+36·3	+5·4
			Original Data—Apparently Reversed Samples				
+165·6	+35·4	+34·0	+26·6732	+4·5	+13·1	+38·2	+6·6
			Normal Dikes and Sills				
+342·4	−33·8	+23·0	+21·0389	+11·2	+9·5	+24·2	+5·2
			Reversed Dikes and Sills				
+167·0	+37·1	+17·0	+15·5816	+11·3	+11·1	+24·1	+6·0
			Normal Flows				
+242·0	−41·8	+14·0	+13·7570	+53·5	+5·5	+11·1	+3·1
			Reversed Flows				
+169·5	+36·6	+20·0	+19·0482	+20·0	+7·5	+18·1	+4·2
			All Normal				
+346·3	−37·1	+37·0	+34·6277	+15·2	+6·3	+20·8	+3·5
			All Reversed				
+168·4	+36·8	+37·0	+34·6242	+15·2	+6·3	+20·8	+3·5
			All Dikes and Sills				
+343·5	−36·9	+40·0	+35·1298	+8·0	+8·5	+28·6	+4·6
			All Flows				
+350·9	−38·8	+34·0	+32·7638	+26·7	+4·9	+15·7	+2·7
			All Flows, Dikes and Sills				
+347·0	−37·9	+74·0	+67·7959	+11·8	+5·0	+23·6	+2·8

The effects of "cleaning" by alternating magnetic fields may be seen by referring to Table 2. The original data could be roughly classified into normal and reversed groups having standard deviations θ_0 of 36°·3 and 38°·2. On cleaning, the standard deviation of each of these groups was reduced to 20°·8.

Because of the geographical extent of the area of collection, part of the scatter of directions of magnetization will be due to differences in longitude and latitude of the sites, and therefore it is perhaps more appropriate to consider the distribution of poles calculated from the directions of magnetization assuming a dipole field. The pole positions given in Table 1 are plotted in Figure 4, different symbols

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being used to denote whether they originate from a normally or reversely magnetized extrusive or intrusive body.

In Creer, Irving & Nairn (1959), it was pointed out that the distribution of a population of directions of magnetization and the distribution of the poles calculated therefrom cannot both be Fisherian, and that if it is assumed that the

Table 3

Average Directions of Magnetization

Site Directions Assigned Unit Weight

D	I	N	R	K	α	θ_0	θ_e
			Normal Dikes and Sills				
+337·9	−34·3	+10·0	+9·4447	+16·2	+12·4	+20·1	+6·7
			Reversed Dikes and Sills				
+170·6	+37·9	+7·0	+6·7405	+23·1	+12·8	+16·8	+6·9
			Normal Flows				
+354·3	−40·4	+5·0	+4·9518	+83·0	+8·4	+8·9	+4·4
			Reversed Flows				
+171·7	+39·0	+8·0	+7·7006	+23·4	+11·7	+16·8	+6·3
			All Normal				
+343·2	−36·7	+15·0	+14·2943	+19·8	+8·8	+18·2	+4·9
			All Reversed				
+171·2	+38·5	+15·0	+14·4400	+25·0	+7·8	+16·2	+4·3
			All Dikes and Sills				
+343·0	−36·0	+17·0	+16·1143	+18·1	+8·6	+19·1	+4·8
			All Flows				
+352·7	−39·6	+13·0	+12·6495	+34·2	+7·2	+13·8	+4·0
			All Flows, Dikes and Sills				
+347·2	−37·6	+30·0	+28·6872	+22·1	+5·7	+17·2	+3·2

Direction of Axial Dipole Field

0° −50° in the south of the area
0° −39° in the north of the area

Present Field Direction

355° −28° in the south of the area
350° −20° in the north of the area

dipole producing the main geomagnetic field undergoes secular oscillations in a random (say Fisherian) manner about the mean dipole for a particular geological period, the resulting directions of magnetization at points on the globe will have oval distributions about their means except at the geographic poles. If the angular standard deviation describing the polar oscillations be θ_0 then the semi-axes of the

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oval describing the distributions of directions of magnetization will depend on the palaeomagnetic colatitude ψ as follows:

$$\delta D = \theta_0/(1+3\cos^2\psi)^{\frac{1}{2}} \quad (1)$$

$$\delta I = 2\theta_0/(1+3\cos^2\psi). \quad (2)$$

Table 4

Average North Pole Position

Site Poles Assigned Unit Weight

Lat°W	Long°N	N	R	K	α	θ_0	θ_s
			Normal Dike Site Poles				
134°·3	+70°·0	10	9·4050	15·1	12°·8	20°·8	6°·9
			Reversed Dike Site Poles				
126°·7	+80°·9	7	6·7681	25·9	12°·1	15°·9	6°·5
			Normal Flow Site Poles				
108°·9	+83°·5	5	4·9770	174·1	5°·8	6°·1	3°·1
			Reversed Flow Site Poles				
109°·7	+81°·5	8	7·7203	25·0	11°·3	16°·2	6°·1
			Normal Site Poles				
130°·6	+74°·8	15	14·2802	19·5	8°·9	18°·4	4°·9
			Reversed Site Poles				
117°·9	+81°·3	15	14·4845	27·2	7°·5	15°·5	4°·2
			Dike Site Poles				
132°·4	+74°·6	17	16·1008	17·8	8°·7	19°·2	4°·8
			Flow Site Poles				
109°·4	+82°·3	13	12·6955	39·4	6°·7	12°·9	3°·7
			All Site Poles				
125°·9	+78°·2	30	28·7123	22·5	5°·7	17°·1	3°·2

Figure 5 consists of centred stereographic plots which have been constructed from Figures 3c and 4, the mean direction being plotted at the centre of the projection, and the other directions having been rotated along appropriate small circles by the required amount. Figure 5a shows the distribution of poles calculated from the mean direction of magnetization at each site. The projection of the cone of standard deviation is drawn in. This is, of course, a circle and has radius 17° (see Table 3). Out of the 30 poles, 19·5, i.e. 65 per cent of them are contained within this circle and this is close to the proportion (63 per cent) which should lie within the standard deviation for a Fisherian distribution (see Creer, Irving & Nairn 1959). Thus, the assumption that the poles are distributed according to Fisher's distribution is justified. In Figure 5b, the distribution of mean directions of magnetization for each site is shown, the mean direction of the set of directions having been plotted at the centre. Using equations (1) and (2), the semi-angles δD and δI of the oval corresponding to the standard deviation circle for the polar

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distribution have been calculated (14°·2 and 24°·2 respectively) and this oval has been drawn. Of the 30 directions, 20·5 (68 per cent) lie within this oval, again in agreement with the assumption that the poles are distributed in a Fisherian manner. However, it cannot be dogmatically stated that the directions of magnetization themselves do not have a Fisherian distribution about their mean, al-

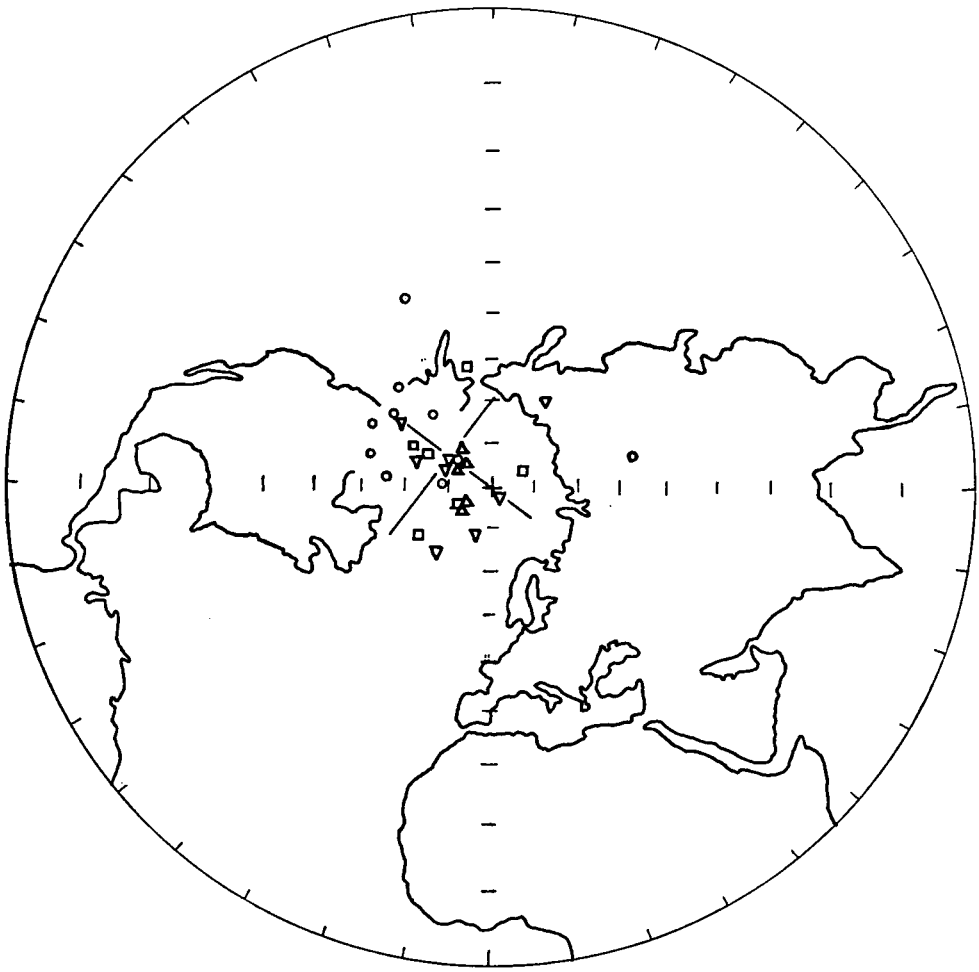


FIG. 4.—Pole positions of site-mean directions of magnetization. Mean pole position (×) is at 126° W, 78° N.

○ Normal Dike □ Reversed Dike

△ Normal Flow ▽ Reversed Flow

though their standard deviation circle, of radius 17°·2 (see Table 2), contains 21·5 out of the 30 points, indicating a rather tighter distribution than the Fisherian one. (Of the observations, 71 per cent lie within the standard deviation as compared with a theoretical 63 per cent for observations distributed according to Fisher. However, there are not enough data to pursue this argument any further.)

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5. Secular variation of the ancient field

Creer (1955) attempted to estimate the amount of secular variation in the remote past from palaeomagnetic observations and suggested that site mean directions of magnetization be used as unit vectors for this purpose. Thus, dispersion of magnetization due to imperfect alignment of the magnetic material in a rock is eliminated. Watson & Irving (1957) have also pointed out that samples from a suitably defined site will have been magnetized at approximately the same time.

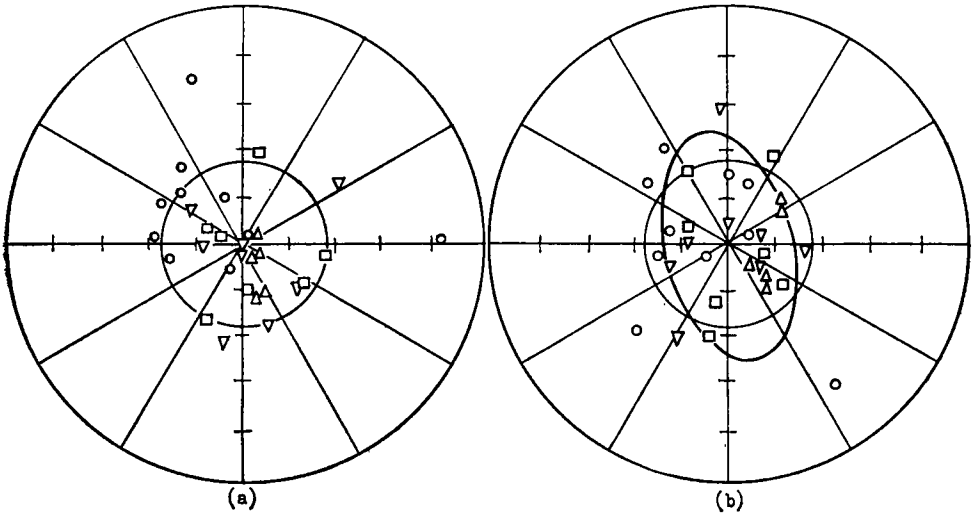


FIG. 5.—Centred stereographic plots:

(a) of site-mean pole positions. The standard deviation circle has radius 17°·1 and contains 19·5 of the 30 poles (65 per cent). (b) of site-mean directions of magnetization. The oval of confidence calculated from the standard deviation circle of Figure 5a using equations (1) and (2) has axes of radii 14°·4 and 24°·6. The standard deviation circle of radius 17°·2 assuming a Fisherian distribution of directions of magnetization is also drawn (Figure 5b). It contains 21·5 of the 30 points (71 per cent). For a Fisherian distribution 63 per cent of the points should lie within the standard deviation.

○ Normal Dike □ Reversed Dike △ Normal Flow ▽ Reversed Flow

Certainly samples from igneous formations can be so chosen in the field, in which case the variation within sites may be largely ascribed to experimental error. The between-sites variation will, however, be due principally to secular variation and subsequent tectonic movement. In undisturbed regions the latter will be comparatively unimportant and this is probably the case as far as the Serra Geral Formation is concerned since the lava flows are almost flat lying today. The time required for the eruption of this huge thickness of flows with intercalated sandstones must inevitably be much longer than times associated with the motions in the core responsible for the secular variation, so that it is reasonable to suppose that there is a full record of the latter stored in the fossil magnetism of this formation. A more pertinent question is whether the 30 site directions are fully representative of this fossil magnetization, the mean really representing the axial dipole palaeofield direction and whether the standard deviation of the distribution of site directions about the mean is fully representative of the secular variation.

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As far as the whole of the data grouped together is concerned, the answer is very probably yes, since samples have been collected from lava flows near the base of the formation and just overlying the Botucatú sandstone, right through to flows near the top just below the overlying Baurú sandstone. Furthermore, the intrusive phase, is also represented in the data. However, for some of the sub-groups (e.g. the normally magnetized flows), the standard deviations may be rather low because the full secular variation may not have been completely sampled within the sub-group.

The standard deviation, θ_0, for the distribution of all 30 of the site directions, 17°, and the corresponding precision $K = 22$, are in reasonable agreement with values of these quantities calculated for various Permian formations (see Creer, Irving & Nairn 1959) and are roughly the same as Creer (1962) has calculated, on certain assumptions, would be produced by the present day secular variation field.

The standard deviation of the data from sites in the basaltic flows (14°) is rather less than that from sites in the diabase dikes and sills (19°). The small difference may be due to the fact that the intrusives are more likely to have been moved slightly since magnetization than the extrusive flows. There may be a contribution due to tectonic movement to the scatter of the dike and sill data.

6. Internal scatter

The statistics θ_0 and K calculated assigning unit weight to sample directions and poles are produced by the sum of the effects of internal and external scatter. Using site directions and poles as unit vectors, θ_0 and K are produced by external effects only. The internal dispersion may be estimated using the following relationship:

$$\frac{1}{K_{\text{total}}} = \frac{1}{K_{\text{int}}} + \frac{1}{K_{\text{ext}}} \tag{3}$$

In Table 5 are listed K values for the various groups of directions of magnetization. The internal scatter of the diabase dikes and sills appears to be considerably greater than that for the basaltic flows. This is what one might expect since basalts are of finer grain size than diabases and are less likely to contain multi-domain magnetite particles which would be less stable than the fine single grained particles. The small difference in K_{ext} for normal and reversed groups probably arises because the ratio of numbers of flow sites and dike sites is different (0·5 and 1·14 respectively for N and R).

7. Intensity of magnetization

Data for a random selection of cores are illustrated in the histograms in Figure 6 in which intensity is plotted on a logarithmic scale along the abscissae. The mean intensity of magnetization (geometric mean) of specimens from the diabase dikes and sills appears to be higher than for specimens of the basaltic flows ($2{\cdot}1 \times 10^{-2}$ and $9{\cdot}6 \times 10^{-3}$ respectively) and furthermore, the range of variation of intensity for the diabase specimens appears to be rather less than for the lava flows.

Subdivision of the data from which Figures 6a and 6b were constructed reveals no apparent difference in intensity between normal and reversely magnetized sub-groups.

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8. Susceptibility

The susceptibility of a randomly selected set of specimens was measured by means of an inductance bridge. The applied field was about 10 Oe and the apparatus was calibrated using a standard specimen containing a solution of manganese sulphate of known strength. This standard specimen was the same shape as the rock specimens. From the histogram in Figure 7 it appears that the volume susceptibility of the basalt is rather lower than that of the diabase, but this suggestion needs investigating further.

Table 5

	Precision K		
	K_{total} (Samples)	K_{ext} (Sites)	K_{int} (Internal)
Normal Flows	53·5	83·0	149
Reversed Flows	20·0	23·4	159
Normal Dikes and Sills	11·2	16·2	36·2
Reversed Dikes and Sills	11.3	23·1	22·1
Flows	26·7	34·2	122
Dikes and Sills	8·0	18·1	14·3
All Normal	15·2	19·8	28·2
All Reversed	15·2	25·0	38·8
All	11·8	22·1	25·3

9. Alternating field demagnetization and thermomagnetic analysis measurements

In Section 4 it was mentioned that the precision of the data was improved after partially demagnetizing the specimens by treatment with alternating magnetic fields using a procedure described previously (Creer (1958, 1959)). The alternating field method of "cleaning" is now widely used and other applications of it are well known (As & Zijderveld (1958), Brynjolfsson (1957), Rimbert (1958)).

Table 6

	Standard Deviation θ_0		
	(Samples)	(Sites)	(Internal)
Normal Flows	11°·1	8°·9	3°·7
Reversed Flows	18°·1	16°·8	3°·3
Normal Dikes and Sills	24°·2	20°·1	13°·5
Reversed Dikes and Sills	24°·1	16°·8	17°·2
Flows	15°·7	13°·8	5°·5
Dikes and Sills	28°·6	19°·1	21°·4
All Normal	20°·8	18°·2	15°·3
All Reversed	20°·8	16°·2	13°·0
All	23°·6	17°·2	16°·1

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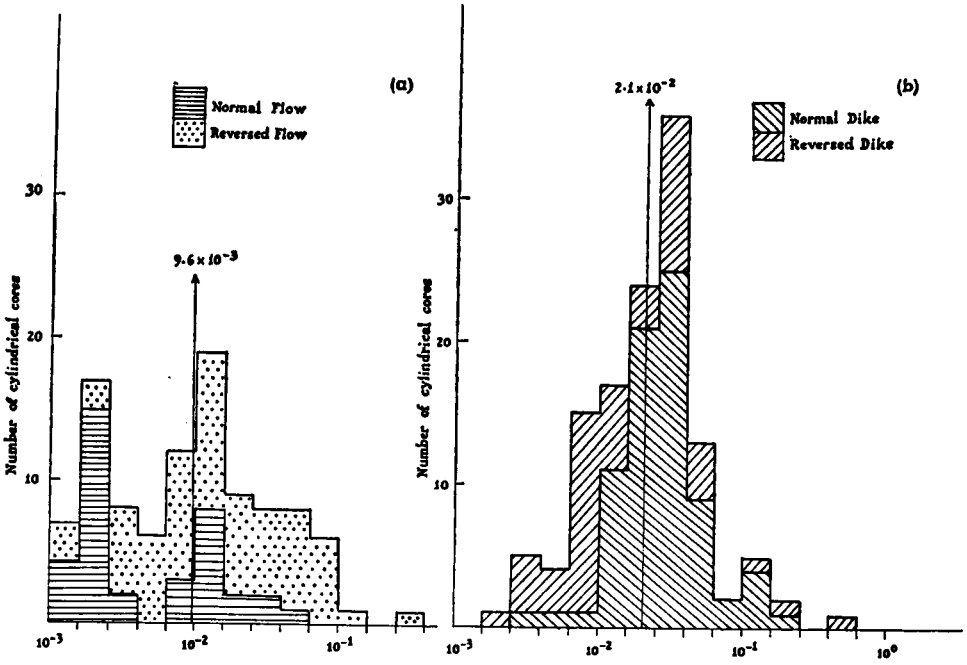


FIG. 6.—Histogram of intensities of magnetization:
(a) of basaltic flows; (b) of diabase dikes and sills.
Volume intensity of magnetization in gauss.

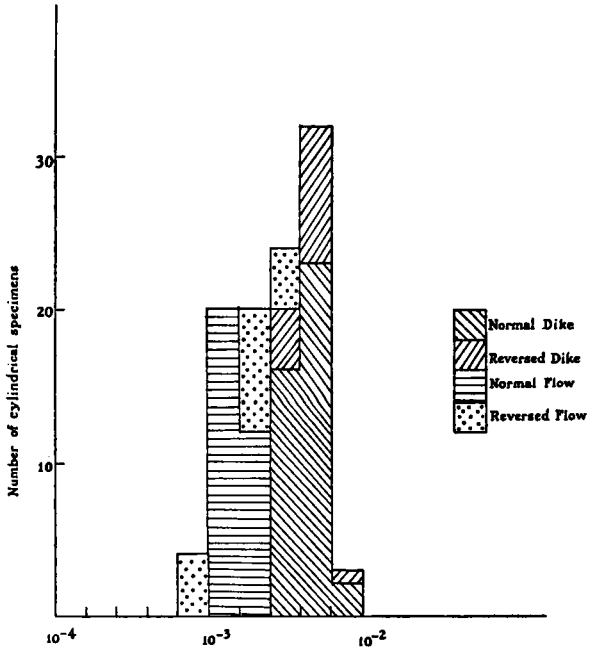


FIG. 7.—Histogram of susceptibilities.
Volume susceptibility (c.g.s.e.m.u.)

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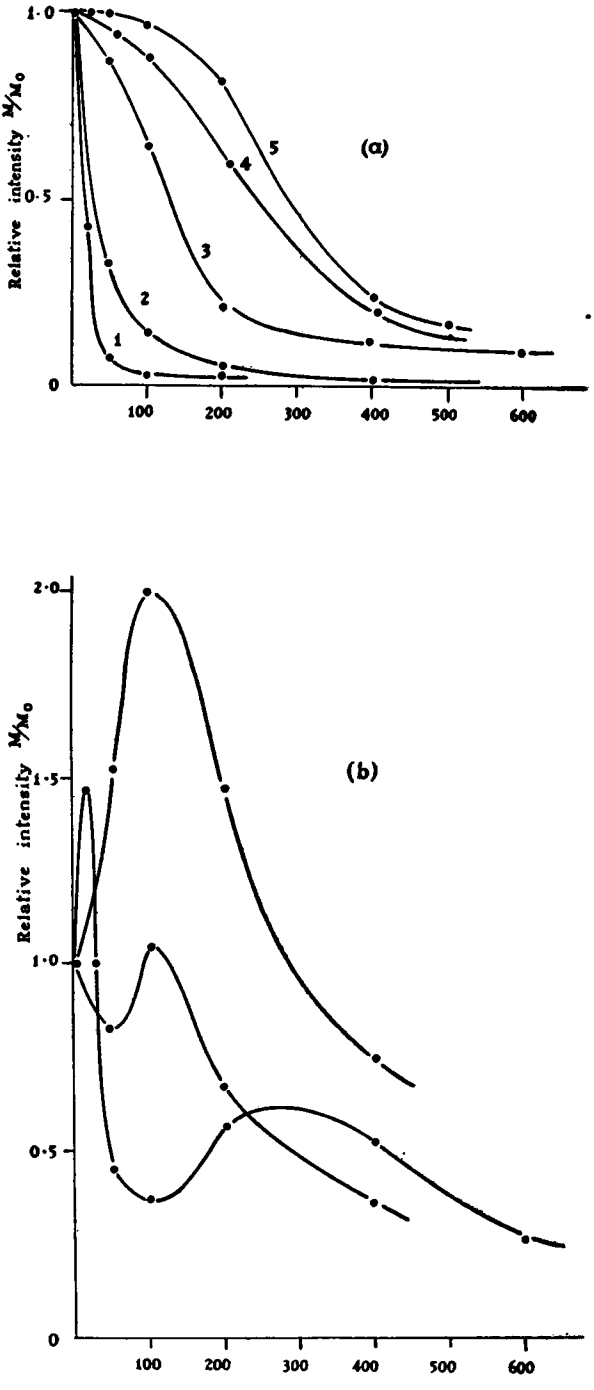


FIG. 8.—Demagnetization curves resulting from treatment by alternating magnetic fields.

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Briefly, the specimen is placed in an alternating field of peak value H_1, the strength of which is gradually and very smoothly reduced to zero. This procedure is carried out in zero steady field so as to prevent the creation of anhysteretic magnetization in the samples which are rotated about two mutually perpendicular axes so as to demagnetize them isotropically. The remanent magnetization is then measured and the process repeated several times in alternating fields $H_2 > H_1$, $H_3 > H_2$, etc.

Demagnetization curves are illustrated in Figures 8a and 8b. The behaviour of the rocks could be broadly classified into three types:

(i) Type 1 showed a marked decrease in intensity after treatment in alternating fields of 50 oersteds (curves 1 and 2, Figure 8a), the final intensity of magnetization being only a few per cent of the initial. The direction of magnetization changed through large angles, even up to 180°. The natural remanence of these rocks was unstable in that repeat measurements after storage in the laboratory in the Earth's field differed from the original by large angles, sometimes 50° or more.

(ii) Type 2 (curves 3, 4 and 5, Figure 8a) were not appreciably affected after treatment in alternating fields of 100 oersteds but several hundreds of oersteds reduced the intensity to not below about 10 per cent of its initial value, after which the direction had rotated through only a few degrees (up to 10°). The mean direction of the several cores from a given hand specimen listed in column II, Table 2 was very often unchanged. The repeatability of measurements of direction of n.r.m. of rocks of this group was good, and could probably be correlated quantitatively with the shape of the demagnetization curves if more work were done.

(iii) Type 3 (Figure 8b) is characterized by an initial increase in intensity as the peak value of the alternating field is increased. The direction of magnetization is rotated through several tens of degrees and the ratio of the final intensity of magnetization to the original varies.

The general features of the three types of curve can be explained in general terms by the existence of two components of magnetization; a hard one which has a demagnetization curve similar to that of a thermoremanent magnetization and a soft component whose demagnetization curve has the characteristics of that of an isothermal remanence. The former component seems to be directed along the axis of the ancient field and to represent a true fossil magnetization. The latter seems to have been built up by the Earth's field while the specimens have been lying in the laboratory. In some cases there seems to be present a viscous component acquired in recent geological times in the direction of the present Earth's field and of rather greater durability than the soft component, but this component is not dominant since there is little evidence of "streaking" of the points on the stereogram (Figure 3a) (see Creer (1957) in which these three components were termed primary, temporary and secondary respectively). In rocks of type 1 it is suggested that the soft component is much stronger than the hard, fossil component; in rocks of type 2, that the fossil component is much stronger than the soft component; whilst in rocks of type 3, that these two components are of approximately the same strength, but pointing in opposite directions. Two almost parallel components of about the same strength would, of course, have a demagnetization curve showing a rapid decrease in intensity (like type 1) and then a more gradual approach to a final steady value.

Thermomagnetic curves for a number of samples were obtained using a Chevallier type magnetic balance. The field in which the measurements were made

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was about 2 000 oersteds. The following points are noted, the results being illustrated in Figure 9:

(i) All samples show a single Curie point of about 580° C, that of magnetite. A sample of pure magnetite gave curve 7.

(ii) No low Curie point minerals are present.

(iii) The sample giving curve 2, which is very closely similar to curve 7 (pure magnetite), possesses an unstable n.r.m. (it was classified as type 1 as a result of alternating field demagnetization having given curve 1, Figure 8a).

(iv) The samples giving more rounded thermomagnetic curves possess n.r.m. more stable to alternating field treatment. The Curie point suggests that the mineral responsible is magnetite or possibly a titaniferous haematite.

(v) Curve 8 is for a sample of baked Tacuarembo sandstone (see Creer 1958), and suggests that magnetite or perhaps titaniferous haematite is also responsible for the n.r.m. of this rock.

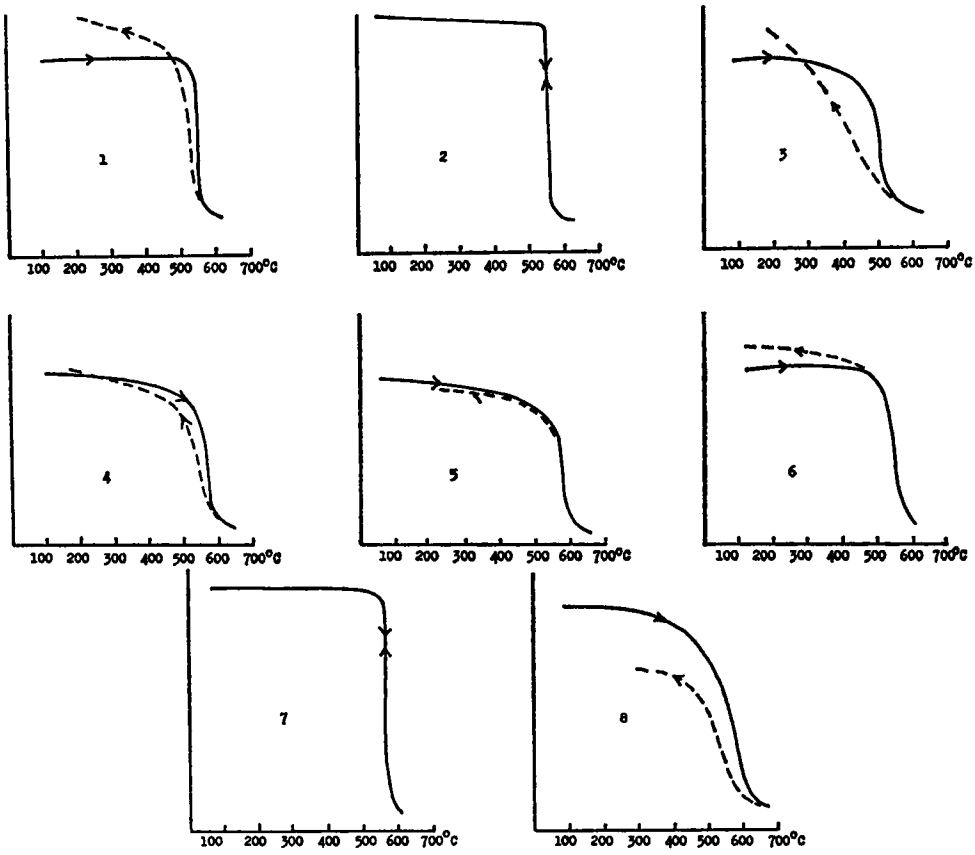


FIG. 9.—Thermomagnetic curves of samples from:

1. Site 10,
2. Site 13,
3. Site 15,
4. Site 22,
5. Site 25,
6. Site 29,
7. Pure magnetite,
8. Baked Tacuarembo sandstone.

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1. Conclusions

(1) The mean direction of magnetization of the Serra Geral Formation is 347°·2 E of present geographic north with upward inclination of 37°·6. This corresponds to a north pole at 125°·9 W longitude and 78°·2 N latitude.

(2) The average direction of magnetization ($D = 347°{\cdot}2$ E, $I = -37°{\cdot}6$) makes an angle of 12° with the average axial dipole field for the area ($D = 0°$, $I = -44°$) (see Table 3). This is about 4 times the standard error θ_e and twice the radius of the 95 per cent error angle α, and so must be considered significant. The angle between the average direction of magnetization and the present field (11°) is also significant.

(3) The angle of inclination of $-37°{\cdot}6$ implies a palaeogeographic latitude of 21° S assuming a dipole field. The present average latitude of the area is 27° S and ranges from about 22° S to 32° S. Hence, the average direction of magnetization, interpreted in terms of dipole field and continental drift theories, implies a southerly movement (away from the equator) of about 6° since the early Mesozoic combined with a counter-clockwise rotation of 13° (the declination D gives the direction of the palaeo-meridian).

(4) The angle between normal and reversed magnetization axes is 6° and that between the directions of magnetization of flows on the one hand and of dikes and sills on the other is 9°. These differences in direction are not considered significant, being about equal to the radii of the circles of confidence drawn at the 95 per cent probability level.

(5) Cleaning by alternating magnetic fields decreased the standard deviation of normal and reversed groups of sample directions from about 37° to 21° (see Table 2).

(6) The standard deviation of site-mean directions of magnetization is about equal to that which would be expected due to secular variation if this produced the same distortions of the dipole field when the rocks were magnetized as it does today. The standard deviation of the distribution of site-pole positions is of the same order of magnitude as that obtained for other formations which have been thoroughly studied.

(7) The internal scatter within sites (see paragraph 6 and Tables 5 and 6) cleaning is greater for dikes and sills than for lavas. This may be because the grain size of magnetite in the lava flows is less than in the dikes and sills since they cooled more rapidly than the latter. Hence, it is likely that there are fewer multi-domain magnetite particles in the lavas than in the dikes.

(8) Thermomagnetic analysis indicates that the mineral responsible for the natural thermoremanence is magnetite or titaniferous haematite. X-ray analysis of the ferromagnetic concentrate indicates magnetite.

(9) The lavas tended to have unstable magnetizations more frequently than the dikes. This is probably due to the fact that samples of the dikes and sills tended to be fresher than those of the lavas, since the former are frequently quarried as roadstone. The latter were collected largely from natural exposures which often tended to be weathered. Instability could also be due to the production of secondary magnetic minerals as a result of weathering. However the presence of these is not revealed by the thermomagnetic curves.

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(10) Instability in fresh samples of dikes and sills seems to be associated with square shaped thermomagnetic curves—(e.g. curve 2, Figure 9).

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